

A Centered-Cauchy Upgrade of Complex-Magnetic Pseudomodes from $h^{-1/7}$ to $h^{-1/4}$

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Lane 0 — candidate for human review

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Status. Candidate full solution, likely valid. Under exactly the hypotheses of source Theorem 1.5, the residual improves from $\exp(-Ch^{-1/7})$ to $\exp(-Ch^{-1/4})$. This fully answers the literal question whether the power $1/7$ is optimal: it is not. No claim is made that $1/4$ is the final sharp power.

1 Source question and result

Krejčíř'ik, Nguyen Duc, and Raymond construct WKB pseudomodes for a two-dimensional Laplacian with a smooth complex magnetic potential. If x^0 belongs to their noncritical set Γ and the magnetic field $\mathbf{B}$ is real analytic at x^0 , their Theorem 1.5 gives

$$\|(\mathcal{L}_{h,\mathbf{A}} - h\mathbf{B}(x^0))u_h\| \leq \exp(-Ch^{-1/7}) \|u_h\|.$$

They explicitly ask whether the power $1/7$ is optimal [1, p. 69 and Theorem 1.5].

setting. Below we provide examples of magnetic potentials for which $\mathbf{A} \neq \emptyset$, these include polynomial and oscillating complex magnetic fields.

In order to prove Theorem 1.5, we use the ideas of the magnetic WKB strategy as developed in [7, 13] in the self-adjoint case to find approximations of eigenfunctions. In our non-self-adjoint development, it is remarkable that the magnetic potential $\mathbf{A}$ does not need to be analytic for this result to hold. This technical outcome is due to the use of a formal gauge, which allows for the local transformation to an appropriate analytic potential at x^0 , see Section 3.1. The power $h^{\frac{1}{7}}$ in (1.6) arises from the transport solutions estimate in Lemma 3.9, which intrinsically reflects the order of the partial derivatives involved in the recursive formula (3.13). However, inspired by the exponential decay $\mathcal{O}(e^{-C/h})$ in the Schrödinger case [11, Theo. 1.1], it is natural to ask whether the power of h in (1.6) is optimal.

We stress that Theorem 1.5 is not covered by [11] (see also [30] and the seminal work [16, Chapter 27]). Indeed, the Weyl symbol of $\mathcal{L}_{h,\mathbf{A}}$ is

$$p(x, \xi) = |\xi - \operatorname{Re} \mathbf{A}|^2 - |\operatorname{Im} \mathbf{A}|^2 - 2i\langle \xi - \operatorname{Re} \mathbf{A}, \operatorname{Im} \mathbf{A} \rangle.$$

Figure 1: Source PDF page 69 (arXiv PDF page 4): the exact optimality question and its connection to the transport estimate.

Theorem 1. *Under the hypotheses of source Theorem 1.5, there are $C, h_0 > 0$ and $u_h \in C_c^\infty(\mathbb{R}^2)$, $0 < h \leq h_0$, such that*

$$\|(\mathcal{L}_{h,\mathbf{A}} - h\mathbf{B}(x^0))u_h\| \leq \exp(-Ch^{-1/4}) \|u_h\|.$$

Consequently the source power $1/7$ is not optimal.

The WKB construction itself is unchanged. The improvement comes from a local Cauchy estimate centered at each evaluation point of the transport recursion.

2 The source recursion and the lost powers

Translate x^0 to zero. Source Theorem 3.2 constructs a phase P and real-analytic amplitudes $(a_j)_{j \geq 0}$ on one neighborhood, with $a_0(0) = 1$ and $a_j(0) = 0$ for $j \geq 1$. The uniform radii fixed in the proof of source Lemma 3.9, especially (3.19), put all their holomorphic complexifications on one common polydisc. In the complex variables (z, w) , write $\tilde{a}_j$ for those complexifications and relabel the source characteristic curve as $w = \omega(z)$.

The exact source recursion (3.13) has the form

$$\begin{aligned} \tilde{a}_{j+1}(z, w) = & J(z, w) \int_{\omega(z)}^w \frac{T_j(z, u)}{2J(z, u)V(z, u)} du \\ & + \tilde{a}_0(z, w) \int_0^z \frac{G_j(u)}{\mathcal{A}_0(u)} du, \end{aligned} \quad (1)$$

where

$$T_j(z, w) = \int_0^1 \partial_w^2 \partial_z \tilde{a}_j(z, \omega(z) + t(w - \omega(z))) dt,$$

and G_j is a bounded analytic linear combination, evaluated at $(u, \omega(u))$, of

$$\partial_z \partial_w^2 \tilde{a}_j, \quad \partial_z^2 \partial_w^2 \tilde{a}_j, \quad \partial_z \partial_w^3 \tilde{a}_j. \quad (2)$$

All the other factors in (1) are fixed holomorphic functions; the source has already shrunk the neighborhood so that the required denominators do not vanish.

The derivative multi-orders in (2) have total orders 3, 4, 4. The source instead uses Cauchy contours centered at the origin. Their denominators carry one extra power in each variable, and the proof then takes the largest separate z - and w -losses simultaneously, producing $(R_1 - |z|)^{-3}(R_2 - |w|)^{-4}$ and hence j^7 per step. The following local form of Cauchy's estimate retains the actual total derivative order.

Lemma 1 (Centered Cauchy estimate). *Let f be holomorphic on the polydisc $\mathcal{P}(r_1, r_2) = \{|z| < r_1, |w| < r_2\}$. Suppose (z, w) has coordinate margins at least $\delta_1, \delta_2 > 0$. Then, for $\alpha, \beta \in \mathbb{N}_0$,*

$$|\partial_z^\alpha \partial_w^\beta f(z, w)| \leq C_{\alpha, \beta} \delta_1^{-\alpha} \delta_2^{-\beta} \|f\|_{r_1, r_2},$$

where $\|f\|_{r_1, r_2} = \sup_{\mathcal{P}(r_1, r_2)} |f|$.

Proof. Apply the two-variable Cauchy formula on circles centered at (z, w) with radii arbitrarily close to (δ_1, δ_2) . Differentiating the Cauchy kernels gives the stated powers (and the harmless factorial constant). $\square$

3 Improved transport growth

Lemma 2. *There are $M, \delta_0 > 0$ such that for every $j \geq 1$ and $x \in D(0, \delta_0) \subset \mathbb{R}^2$,*

$$|a_j(x)| + |\nabla a_j(x)| + |\Delta a_j(x)| \leq M^{j+1} j^{4j}. \quad (3)$$

Proof. Choose $R_1, R_2 > 0$ so that the closed polydisc with radii $(2R_1, 2R_2)$ lies in the common holomorphic neighborhood of all data in (1). As in the source proof, after shrinking we have

$$|\omega(z)| \leq C_0 |z|, \quad C_0 R_1 < R_2, \quad (4)$$

and every fixed coefficient in (1) is uniformly bounded.

Fix $n \geq 1$. For $k = 0, \dots, n$, set

$$r_{1,k} = \left(2 - \frac{k}{n}\right) R_1, \quad r_{2,k} = \left(2 - \frac{k}{n}\right) R_2.$$

The coordinate gaps between consecutive polydiscs are exactly R_1/n and R_2/n .

Take $(z, w) \in \mathcal{P}(r_{1,k+1}, r_{2,k+1})$. Both endpoints $\omega(z)$ and w lie in the w -disc of radius $r_{2,k+1}$, by (4); hence their entire joining segment does too. If u lies on the segment from 0 to z , then

$$|u| \leq |z| < r_{1,k+1}, \quad |\omega(u)| \leq C_0|u| < r_{2,k+1}.$$

Thus every point where a derivative of $\tilde{a}_k$ is evaluated in (1) lies in the $(k+1)$ -st polydisc and has margins at least $R_1/n, R_2/n$ inside the k -th polydisc.

Lemma 1 and the exhaustive derivative list (2) now give

$$\|T_k\|_{r_{1,k+1}, r_{2,k+1}} \leq Cn^3 \|\tilde{a}_k\|_{r_{1,k}, r_{2,k}}, \quad \sup |G_k| \leq Cn^4 \|\tilde{a}_k\|_{r_{1,k}, r_{2,k}}.$$

The two path lengths in (1) and all fixed coefficients are bounded independently of k, n . Therefore

$$\|\tilde{a}_{k+1}\|_{r_{1,k+1}, r_{2,k+1}} \leq Cn^4 \|\tilde{a}_k\|_{r_{1,k}, r_{2,k}}.$$

Multiplication for $k = 0, \dots, n-1$ yields

$$\|\tilde{a}_n\|_{R_1, R_2} \leq (Cn^4)^n \|\tilde{a}_0\|_{2R_1, 2R_2}.$$

A final application of Lemma 1 across the fixed margins from (R_1, R_2) to $(R_1/2, R_2/2)$ gives the same growth for the first derivatives and for $4\partial_z\partial_w\tilde{a}_n = \widetilde{\Delta}a_n$. Enlarging one constant and restricting to a fixed real disk proves (3). $\square$

4 Optimal truncation and proof of Theorem 1

Let M be as in Lemma 2, and for small $h > 0$ put

$$N(h) = \left\lfloor (eMh)^{-1/4} \right\rfloor.$$

For $1 \leq j \leq N(h)$,

$$Mhj^4 \leq e^{-1}, \quad h^j M^{j+1} j^{4j} \leq Me^{-j}.$$

Consequently the truncated amplitude and its first derivatives are uniformly bounded. Since $a_j(0) = 0$ for $j \geq 1$, integration of the gradient along $[0, x]$ also gives

$$\sum_{j=1}^{N(h)} h^j |a_j(x)| \leq C|x|.$$

Together with $a_0(0) = 1$, this supplies a uniform lower bound on the amplitude near zero.

Use the source phase P and a fixed cutoff χ supported in the disk where (3) holds:

$$u_h = \chi e^{-P/h} \sum_{j=0}^{N(h)} h^j a_j.$$

The positive-definite quadratic expansion of $\operatorname{Re} P$ implies

$$\|u_h\| \geq ch^{1/2}; \tag{5}$$

indeed, integrate on a real ball of radius comparable to $\sqrt{h}$.

The source transport identities cancel every interior power through $h^{N(h)+1}$. Lemma 2 bounds the remaining coefficient by

$$\begin{aligned} h^{N+2} |\Delta a_N| &\leq Mh^2 (MhN^4)^N \\ &\leq Mh^2 e^{-N} \leq \exp(-ch^{-1/4}) \end{aligned}$$

after decreasing c and h_0 . Its Gaussian L^2 norm is $O(h^{1/2})$, which matches (5). On the support of $\nabla\chi$, $\operatorname{Re} P$ has a fixed positive lower bound; the unchanged commutator term is therefore $O(e^{-c_0/h})$ up to polynomial factors. After division by (5), this is smaller than $e^{-ch^{-1/4}}$. We have used the source's gauge-restored identity (3.14), whose phase is already $P = S + i\theta$, so no additional gauge factor has to be estimated here. Combining the two remainders proves Theorem 1.

5 Verification, novelty, and scope

The proof was checked in nine independent pieces: common analyticity, characteristic-curve geometry, centered Cauchy margins, exhaustive derivative orders, the one-step recursion, iteration, fixed-margin derivative recovery, optimal truncation, and relative L^2 normalization. A deterministic script independently verifies the multi-index and truncation bookkeeping.

A bounded search on 11 August 2026 covered the local run and full-source indexes, the exact title/id and $h^{1/7}$ wording, close official arXiv searches for a $1/4$ improvement and local-Cauchy WKB estimates, the September 2025 arXiv revision, and the January 2026 publication record. No prior exponent improvement was found. The argument proves only that $1/7$ is not optimal; whether $1/4$ can be improved, possibly all the way to 1, remains open here.

Human review focus. Verify the path-containment paragraph in Lemma 2, compare (2) term by term with source equation (3.13), and check that the final Gaussian norm comparison is performed before polynomial factors are absorbed into the exponential.

References

- [1] D. Krejčířik, T. Nguyen Duc, and N. Raymond, *The Laplacian with complex magnetic fields*, SIAM J. Math. Anal. 58 (2026), 66–91; [doi:10.1137/24M1703628](https://doi.org/10.1137/24M1703628); [arXiv:2410.01377](https://arxiv.org/abs/2410.01377).